

- a. Using the quadratic regression model, $PRICE = \alpha_1 + \alpha_2 SOFT^2 + e$, test the hypothesis that the marginal effect on expected house price of increasing the size of a 2000 square foot house by 100 square feet is less than or equal to \$13000 against the alternative that the marginal effect will be greater than \$13000. Use the 5% level of significance. Clearly state the test statistic used, the rejection region, and the test p -value. What do you conclude?

```
rm(list=ls())
library(POE5Rdata)
#####
#problem3.23
alpha = 0.05
mod1 = lm(price ~ I(sqft^2), data = collegetown)
b2 = coef(mod1)[[2]]
df = df.residual(mod1)

# H0 : 40*ALPHA2 <= 13 , H1 : 40*ALPHA2 > 13
#->H0 : ALPHA2 <= 0.325 , H1 : ALPHA2 > 0.325
tc = qt(1-alpha, df)
smod1 = summary(mod1)
se_b2 = coef(smod1)[2,2]
t_star = (b2 - 0.325)/se_b2

cat(
  "3.23(a)", "\n",
  "test statistic is: t = (alpha2_hat - 0.325)/se(alpha2_hat)", "\n",
  "rejection region: tc > ", tc, "\n",
  "test t-value : ", t_star, "\n",
  "hence t* is falls in the non-rejection region", "\n",
  sep=""
)
```

虛無假設 $H_0: 40 \cdot \text{ALPHA2} \leq 13$

對立假設 $H_1: 40 \cdot \text{ALPHA2} > 13$

t 值為 -26.72875

拒絕域為 t 值 大於 1.647919

p-value: 近乎為 1

結論：無法拒絕虛無假說 (Fail to reject H_0)

- b. Using the quadratic regression model in part (a), test the hypothesis that the marginal effect on expected house price of increasing the size of a 4000 square foot house by 100 square feet is less than or equal to \$13000 against the alternative that the marginal effect will be greater than \$13000.

#3.23(b)

```
# H0 : 80*ALPHA2 <= 13 , H1 : 80*ALPHA2 > 13
#->H0 : ALPHA2 <= 0.1625 , H1 : ALPHA2 > 0.1625
t_star_b = (b2 - 0.1625)/se_b2

cat(
  "3.23(b)", "\n",
  "test statistic is: t = (alpha2_hat - 0.1625)/se(alpha2_hat)", "\n",
  "rejection region: tc > ", tc, "\n",
  "test t-value : ", t_star_b, "\n",
  "hence t* is falls in the non-rejection region", "\n",
  sep=""
)
```

虛無假設 $H_0: 80 \cdot \text{ALPHA2} \leq 13$

對立假設 $H_1: 80 \cdot \text{ALPHA2} > 13$

t 值為 4.189467

拒絕域為 t 值 大於 1.647919

p-value: 1.65395e-05

結論：拒絕虛無假說 (Reject H_0)，數據顯示 4000 平方英尺房子的邊際效應超過 \$13,000

- c. Using the quadratic regression model in part (a), estimate the expected price $E(\text{PRICE}|\text{SQFT}) = \alpha_1 + \alpha_2 \text{SQFT}^2$ for a house of 2000 square feet. Construct a 95% interval estimate of the expected price. Describe your interval estimate to a general audience.

```
#3.23(c)
# E(price|sqft=20)_hat = alpha1_hat + 400 * alpha2_hat
b1 = coef(mod1)[[1]]
e_hat = b1 + 400 * b2
se_l = sqrt(vcov(mod1)[1,1] + 160000*vcov(mod1)[2, 2] + 800*vcov(mod1)[1,2])
tc_c = qt(1-alpha/2, df)
lowb = e_hat - tc_c*se_l
upb = e_hat + tc_c*se_l

cat(
  "3.23(c)", "\n",
  "expected price is ", e_hat, "\n",
  "95% CI of the expected price: [", lowb, ",", upb, "]", "\n",
  "In repeated sampling, 95% intervals constructed this way will contain true value", "\n",
  "of expected price", "\n",
  sep=""
)
```

Expected price is 167.3735 and the 95% CI of the expected price is [158.0481, 176.6988].

Meaning: In repeated sampling, 95% intervals constructed this way will contain true value of expected price

- d. Locate houses in the sample with 2000 square feet of living area. Calculate the sample mean (average) of their selling prices. Is the sample average of the selling price for houses with $SQFT = 20$ compatible with the result in part (c)? Explain.

```
)#3.23(d)
d <- collegetown$sqft == 20
d_sqft20 <- collegetown[d, ]
s_avg = mean(d_sqft20$price)

cat(
  "3.23(d)", "\n",
  "In the sample there are 3 houses with 2000 square feet", "\n",
  "and sold ", d_sqft20$price[1], ", ", d_sqft20$price[2], ", ", d_sqft20$price[3], "\n",
  "The sample mean is ", s_avg, "\n",
  "and it is inside the interval we estimate", "\n",
  sep=""
)
```

In the sample there are 3 houses with 2000 square feet, and sold 138,169,183. As a result, the sample mean is 163.3333 and it is inside the interval we estimate.